

Problem A. Another Searching Problem

Input file: standard input
 Output file: standard output
 Time limit: 3 seconds
 Memory limit: 512 megabytes

You are studying the following graph algorithm. The graph has n vertices, numbered $1, 2, \dots, n$, and the neighbors of each vertex are visited in increasing order of their labels.

`dp()`:

```

Let q be an empty queue
Initialize id[1..n] to n+1
counter = 0, timer = 0
Push 1 into q

While q is not empty:
  Pop the front element of q into u
  If id[u] equals n+1:
    timer = timer + 1
    id[u] = timer

    Enumerate all neighbors v of u in increasing order:
      counter = counter + 1
      If id[v] > id[u]:
        Push v into q

Return counter

```

Note that the same vertex may be pushed into and popped from the queue multiple times.

Given an integer k , construct a simple graph with n vertices and m undirected edges such that the algorithm above returns k . If no such graph exists, report that there is no solution.

For each test case, the construction must satisfy $1 \leq n \leq 200, 1 \leq m \leq 300$.

Input

The first line contains an integer t ($1 \leq t \leq 2 \times 10^4$), the number of test cases.

Each of the next t lines contains one integer k ($1 \leq k \leq 10^{18}$).

Output

Output the answer for each test case in order.

If no valid construction exists, output a single line containing the integer -1 .

Otherwise, first output a line containing two integers n and m , the number of vertices and edges in the constructed graph, respectively. Then output m lines, each containing two integers u, v , representing an undirected edge connecting vertices u and v .

The output graph must be simple: no self-loops and no multiple edges are allowed. The order of the edges in the output is arbitrary; the algorithm always visits neighbors in increasing order of their labels. The graph does not need to be connected.

If there are multiple valid constructions, output any of them.

Example

standard input	standard output
3	-1
1	2 1
2	1 2
8	3 3
	1 2
	1 3
	2 3

Note

For the first test case, $k = 1$ has no solution. If vertex 1 has no neighbors, the algorithm returns 0; otherwise, when vertex 1 is popped for the first time, at least one neighbor is pushed into the queue, and that neighbor, when later popped, will visit at least one edge, so the returned value is at least 2.

For the second test case, the graph has only one edge $(1, 2)$. The algorithm first visits this edge from vertex 1, and then visits it again from vertex 2, so the return value is 2.

For the third test case, the output is a triangle. The algorithm pops vertices 1, 2, and 3 in order, and then vertex 3 is popped once more. Each pop visits 2 edges, so the return value is 8.

Problem B. Bingo Game

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 512 megabytes

Given an $n \times n$ grid. There are m cells in the grid that have already been lit up. The i -th lit cell is located at row x_i and column y_i .

You may continue to light up any cells that have not yet been lit.

You need to determine the minimum number of additional cells that must be lit so that there exists at least one row or one column in the grid whose n cells are all lit.

Input

The first line contains two integers n, m ($1 \leq n, m \leq 10^5$).

The next m lines each contain two integers x_i, y_i ($1 \leq x_i, y_i \leq n$).

It is guaranteed that all given cell coordinates are pairwise distinct.

Output

Output an integer representing the minimum number of additional cells that need to be lit.

Examples

standard input	standard output
4 5 1 1 1 3 2 2 3 2 4 2	1
3 3 2 1 2 2 2 3	0

Note

In sample 1, the cells in rows 2, 3, 4 of column 2 have already been lit. It is enough to light cell (1, 2), and then column 2 will be completely lit.

In sample 2, all cells in row 2 have already been lit, so no additional cells need to be lit.

Problem C. Drifting Toward a Promised Sky 2

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 512 megabytes

“Even when we’re not looking up at the stars, they’re always there, watching over us.”

Countless jellyfish that once accompanied you have turned into specks of starlight, slowly ascending into the boundless night sky. They can be represented by n points on an infinitely long number line. Initially, the i -th point is located at a_i . Multiple points may occupy the same position.

Index wants to perform at most $10^{218105633}$ evolutions on these points (the number of evolutions may be 0). In one evolution, she will:

1. choose a position x that currently contains at least two points;
2. decrease the number of points at x by 2;
3. increase the number of points at $x - 1$ and $x + 1$ by 1 each.

Index wants to know the minimum possible value of the mex of the set of positions occupied by all points after the operations end.

Note: For an integer set S , $\text{mex}(S)$ is defined as the smallest non-negative integer not in S . For example, $\text{mex}\{1, 2, 3\} = 0$, and $\text{mex}\{0, 2, 5\} = 1$.

Input

This problem contains multiple test cases. The first line contains a positive integer t ($1 \leq t \leq 10$), representing the number of test cases.

For each test case, the first line contains a positive integer n ($1 \leq n \leq 5 \cdot 10^5$), and the second line contains n integers $a_1, \dots, a_n$ ($0 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $5 \cdot 10^5$.

Output

For each test case, output one integer on a separate line, representing the minimum achievable mex.

Example

standard input	standard output
2	4
8	0
6 1 6 5 3 2 0 6	
4	
0 0 1 0	

Note

For the first test case, it can be proved that the initial mex is already the minimum achievable mex.

For the second test case, the following operations can be performed in order:

$$\begin{aligned} [0, 0, 0, 1] &\rightarrow [-1, 0, 1, 1], \\ [-1, 0, 1, 1] &\rightarrow [-1, 0, 0, 2], \\ [-1, 0, 0, 2] &\rightarrow [-1, -1, 1, 2]. \end{aligned}$$

Here, red indicates the two points chosen in the current operation, and blue indicates the newly created points.

At this point, the mex of the set is 0. Clearly, the mex cannot be smaller than 0, so the minimum value is 0.

Problem D. Escape Root

Input file: standard input
Output file: standard output
Time limit: 2 seconds
Memory limit: 512 megabytes

You are given a tree with n vertices, numbered from 1 to n , rooted at vertex 1. Each edge has length 1. There are m people. The i -th person appears at vertex x_i at time s_i seconds, and immediately starts moving toward vertex 1 along the shortest path at a speed of 1 unit per second.

If, at any moment, at least two people are at the same position, then all of those people disappear simultaneously. A position can be either a vertex or the interior of an edge.

If a person reaches vertex 1 without meeting anyone else, then they successfully escape and are immediately removed from the tree.

Determine which people successfully escape.

Input

The first line contains two integers n, m ($1 \leq n, m \leq 2 \cdot 10^5$).

The next $n - 1$ lines each contain two integers u, v ($1 \leq u, v \leq n$, $u \neq v$), indicating that there is an edge connecting u and v in the tree.

The next m lines each contain two integers x_i, s_i ($1 \leq x_i \leq n$, $0 \leq s_i \leq 10^9$), indicating the vertex where the i -th person appears and the time when they appear.

Output

Output a 01 string of length m .

The i -th character should be 1 if the i -th person successfully escapes, and 0 if they disappear.

Examples

standard input	standard output
5 4 1 2 1 3 2 4 2 5 4 0 5 0 3 0 2 3	0011
4 5 1 2 2 3 2 4 3 0 4 0 2 1 1 2 4 5	00011

standard input	standard output
2 2 1 2 2 0 1 1	00

Note

In sample 1, the first and second people start from vertices 4 and 5, respectively. They both reach vertex 2 at time 1, so they disappear simultaneously.

The third person starts from vertex 3, reaches the root at time 1, and escapes. The fourth person starts from vertex 2 at time 3, reaches the root at time 4, and escapes.

In sample 2, the first three people are all at vertex 2 at time 1, so they all disappear. Although the fourth person appears at the root at time 2, the first three people have already disappeared earlier, so the fourth person can successfully escape.

In sample 3, the first person reaches the root at time 1, while the second person appears at the root at time 1. They meet and disappear simultaneously.

Problem E. Grand Mex 2

Input file: standard input
Output file: standard output
Time limit: 2.5 seconds
Memory limit: 512 megabytes

There are n non-negative integers $a_1, a_2, \dots, a_n$, all initially equal to 0.

You are given a tree with n vertices. For each edge (u, v) in the tree, you must choose exactly one of the following two operations:

- set $a_u \leftarrow a_u + 1$;
- set $a_v \leftarrow a_v + 1$.

Each edge is operated on exactly once, so there are 2^{n-1} possible choices. Different choices are counted separately, even if they produce the same final array.

The mex of a non-negative integer array is the smallest non-negative integer that does not appear in the array. For example, $\text{mex}\{1, 2, 3\} = 0$, $\text{mex}\{0, 2, 5\} = 1$.

Find the sum of the mex values of the final arrays produced by all choices. Output the answer modulo 998244353.

Input

The first line contains an integer n ($2 \leq n \leq 150$).

The next $n - 1$ lines each contain two integers u, v ($1 \leq u, v \leq n$, $u \neq v$), representing an edge in the tree. It is guaranteed that the given edges form a tree.

Output

Output an integer: the sum of the mex values of the final arrays produced by all 2^{n-1} choices, modulo 998244353.

Examples

standard input	standard output
3 2 1 2 3	7
10 10 5 8 9 1 8 2 9 7 3 3 4 8 5 1 4 6 8	1566

Note

There are four choices in total. One of them produces an array whose mex is 1, while the other three produce arrays whose mex is 2. Therefore, the answer is $1 + 2 + 2 + 2 = 7$.

Problem F. Light the Lamp

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 512 megabytes

On an $n \times n$ grid, there is a lamp in every cell. At time 0, exactly one lamp is lit in each row and each column. More specifically, the lamp in row i and column p_i is lit, where p is a permutation of length n .

After that, time advances in whole seconds. For every lamp that is not yet lit, if at time $t - 1$ at least two of its edge-adjacent neighboring cells contain lit lamps, then this lamp becomes lit at time t . A lamp that is already lit never turns off.

You need to determine whether all lamps will eventually become lit. If all lamps do become lit, you also need to find the earliest time when this happens.

Input

The first line contains an integer n ($2 \leq n \leq 5 \cdot 10^5$).

The second line contains n pairwise distinct integers $p_1, p_2, \dots, p_n$ ($1 \leq p_i \leq n$), meaning that initially the lamp in row i and column p_i is lit.

Output

If at least one lamp will never become lit, output -1 .

Otherwise, output an integer denoting the earliest time when all lamps become lit.

Examples

standard input	standard output
4 1 2 3 4	3
4 2 1 4 3	4
4 2 4 1 3	-1

Note

In sample 1, the initially lit lamps are located on the main diagonal. Every second, the lit area expands by one cell to both sides, and the two farthest corners become lit at time 3.

In sample 3, the process stops too early, so it is impossible to light the entire grid.

Problem G. Passing Game

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 512 megabytes

There are n people standing in a circle in clockwise order, numbered $1, 2, \dots, n$. The previous person of person 1 is person n , and the next person of person n is person 1.

In one pass, the current ball holder must pass the ball to either their previous person or their next person.

Person i can pass the ball at most a_i times; the number of times they receive the ball is unlimited. If a person receives the ball and the game ends immediately, then this receive does not consume one of their passing opportunities.

You may choose the initial ball holder arbitrarily, and you may also end the game after any pass. Please find the maximum number of passes that can be made.

Input

5000 The first line contains an integer n ($3 \leq n \leq 5 \cdot 10^5$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$).

Output

Output an integer representing the maximum number of passes.

Examples

standard input	standard output
3 2 2 2	6
5 2 0 3 1 0	3
5 1 1 1 1 3	6

Note

In sample 1, the passers can be in the order 1, 3, 2, 1, 3, 2, so each person passes exactly 2 times, and the total number of passes is 6.

In sample 2, people numbered 2 and 5 cannot pass the ball, so the passing process cannot go through them. One optimal solution is to have the passers in the order 3, 4, 3, for a total of 3 passes.

In sample 3, one optimal solution is to have the passers in the order 2, 3, 4, 5, 1, 5, for a total of 6 passes.

Problem H. Set

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 512 megabytes

Given a sequence of n pairwise distinct integers $a_1, a_2, \dots, a_n$, let the initial set be $S = \{a_1, a_2, \dots, a_n\}$.

You may repeatedly perform the following operation:

1. Choose any element x from S .
2. Split x into two positive integers u, v such that $u + v = x$.
3. Apply symmetric difference operations to S in order: $\oplus x, \oplus u, \oplus v$.

Here, the set operation $S \oplus y$ is defined as follows:

- If $y \in S$, remove y from S ;
- If $y \notin S$, add y to S .

How many different sets S containing **only one element** can be obtained after performing the operation zero or more times?

Input

The first line contains a positive integer n ($1 \leq n \leq 10^5$).

The second line contains n positive integers $a_1, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

Output

Output one line containing an integer representing the answer.

Examples

standard input	standard output
2 2 4	2
3 2 3 4	2

Note

In sample 1, choosing $x = 4, u = 2, v = 2$ can make $S = \{2\}$, and choosing $x = 2, u = 1, v = 1$ can make $S = \{4\}$. It can be proved that no other sets S containing only one element exist.

Problem I. Slay the Spire

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 512 megabytes

You are in a battle. The enemy initially has x health points and an initial poison stack of 0.

In each turn, you must choose one of the following two actions:

1. **Apply Poison:** increase the enemy's poison stack by a ;
2. **Normal Attack:** immediately deal b damage to the enemy.

After your action ends, if the enemy's poison stack is greater than 0, poison is resolved immediately:

1. deal damage to the enemy equal to the current poison stack;
2. then decrease the enemy's poison stack by 1.

The poison stack is not cleared between turns. In particular, poison applied in the current turn will also take effect in the poison resolution after your action ends in this turn.

When the enemy's health becomes less than or equal to 0, the enemy is killed and the battle ends immediately. The turn in which the kill happens is counted as well.

Your task is to find the minimum number of turns needed to kill the enemy under an optimal strategy.

Input

The input contains one line with three integers x, a, b ($1 \leq x, a, b \leq 10^9$), representing the enemy's initial health, the amount of poison applied each time, and the damage dealt by a normal attack.

Output

Output one integer, the minimum number of turns required to kill the enemy.

Examples

standard input	standard output
10 3 4	3
100 5 1	7

Note

In sample 1, one optimal strategy is as follows:

- On turn 1, apply 3 stacks of poison, which deals 3 damage, leaving 2 poison stacks;
- On turn 2, normal attack deals 4 damage, and poison then deals 2 damage, leaving 1 poison stack;
- On turn 3, normal attack, killing the enemy.

In the first 2 turns, at most 9 damage can be dealt, so the answer is 3.

In sample 2, the maximum damage in the first 6 turns is 90. This can be achieved by applying poison every turn, in which case the poison damage is 5, 9, 13, 17, 21, 25 in order; if poison is applied again on turn 7, it will deal another 29 damage, so the enemy can be killed on turn 7.

Problem J. Start to the End

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 512 megabytes

Given n, m, a, b , count how many directed bipartite graphs satisfy the following conditions:

- The left part consists of n labeled vertices, and the right part consists of m labeled vertices.
- The outdegree of every vertex is at most 1. On the left part, there are at most a vertices with outdegree 0; on the right part, there are at most b vertices with outdegree 0.
- Starting from any vertex with nonzero outdegree, it is impossible to reach a vertex with outdegree 0.

Since the answer may be very large, you only need to output the result modulo 2^{64} .

There are multiple test cases, t in total.

Input

This problem contains multiple test cases. The first line contains a positive integer t ($1 \leq t \leq 10^3$), denoting the number of test cases.

For each test case, one line contains four integers n, m, a, b ($0 \leq a \leq n \leq 10^7, 0 \leq b \leq m \leq 10^7$).

Output

For each test case, output one line containing an integer, the answer modulo 2^{64} .

Example

standard input	standard output
6	1
1 1 0 0	2
1 1 1 1	16
2 2 0 0	28
2 2 1 1	150
2 3 1 2	1369
3 3 3 3	

Note

For the first test case, there is one vertex on each side, and neither of them can have outdegree 0. Therefore, they must be connected to each other, so there is only 1 valid directed bipartite graph.

For the second test case, both vertices may have no outgoing edges. There are two valid cases: either both vertices have no outgoing edges, or they are connected to each other. Therefore, the answer is 2.

Problem K. Tree Reflection

Input file: standard input
Output file: standard output
Time limit: 8 seconds
Memory limit: 512 megabytes

Given an integer n and an array a of n positive integers.

There is an integer x , initially equal to 0. You need to perform n random operations in order.

You are also given n rooted outward trees $T_1, T_2, \dots, T_n$. For each $1 \leq i \leq n$, the vertex set of T_i is $\{0, 1, \dots, a_i\}$, the root is 0, and the sequence $0, 1, \dots, a_i$ is a topological order of T_i . In other words, if $u > 0$, then the parent $f_{i,u}$ of u in T_i satisfies $0 \leq f_{i,u} < u$.

You are also given n integers $b_1, b_2, \dots, b_n$. In the i -th operation:

- with probability $\frac{b_i}{998244353}$, increase x by a_i ;
- with probability $1 - \frac{b_i}{998244353}$, decrease x by a_i .

If after this choice $x < 0$, let $u = -x$, and then x is immediately replaced by $f_{i,u}$. Since before the operation x is always non-negative, once $x < 0$ occurs, we must have $1 \leq u \leq a_i$, so the parent above always exists.

Please find the expected value of x after all n operations are finished.

Since the answer is a rational number, you need to output its value modulo 998244353. More specifically, if the answer is the irreducible fraction $\frac{A}{B}$, you need to output $A \cdot B^{-1} \bmod 998244353$. It can be proved that B is coprime with 998244353.

Input

The first line contains an integer n ($1 \leq n \leq 5 \cdot 10^5$).

The second line contains n positive integers $a_1, a_2, \dots, a_n$ ($1 \leq \sum_{i=1}^n a_i \leq 5 \cdot 10^5$).

The third line contains n integers $b_1, b_2, \dots, b_n$ ($0 \leq b_i \leq 998244353$).

The next n lines describe the trees. The i -th line contains a_i integers $f_{i,1}, f_{i,2}, \dots, f_{i,a_i}$ ($0 \leq f_{i,u} < u$), representing the parent of each non-root vertex in tree T_i .

Output

Output one line containing an integer, the expected value of x modulo 998244353.

Example

standard input	standard output
2 2 1 499122177 499122177 0 1 0	499122178

Note

After the first operation, x is 2 with probability $\frac{1}{2}$ and 1 with probability $\frac{1}{2}$.

The probability of choosing addition in the second operation is also $\frac{1}{2}$. Finally:

- $x = 3$ with probability $\frac{1}{4}$;

- $x = 1$ with probability $\frac{1}{4}$;
- $x = 2$ with probability $\frac{1}{4}$;
- $x = 0$ with probability $\frac{1}{4}$.

Therefore, the expected value is $3 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} = \frac{3}{2}$, which equals 499122178 modulo 998244353.

Problem L. XOR Team

Input file: standard input
Output file: standard output
Time limit: 4 seconds
Memory limit: 512 megabytes

There is an initial 01 sequence b of length 10^{18} , all of whose elements are 0.

There is a sequence a of length $2 \cdot n$. You need to divide it into n pairs x_i, y_i , with the condition that $x_i \leq y_i$. Then perform n operations on sequence b : in each operation, XOR all numbers from x_i to y_i with 1.

You need to find the minimum possible sum of sequence b after the operations, over all possible pairings.

Input

This problem has multiple test cases. The first line contains a positive integer t ($1 \leq t \leq 10^5$), representing the number of test cases.

For each test case, the first line contains a positive integer n ($1 \leq n \leq 10^6$). The second line contains $2n$ positive integers $a_1, a_2, \dots, a_{2n}$ ($1 \leq a_i \leq 10^{18}$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^6$.

Output

For each test case, output a non-negative integer representing the minimum possible sum of sequence b after the operations, over all possible pairings.

Example

standard input	standard output
3	4
1	2
2 5	1
2	
1 4 2 3	
3	
1 5 5 1 3 5	

Note

For the first test case, there is only one way to pair them, namely pairing 2 and 5. The corresponding operation interval is $[2, 5]$. After the operation, the values at positions 2, 3, 4, 5 are all 1, so the answer is 4.

For the second test case, one optimal pairing is (1, 4) and (2, 3). After the operations, the values at positions 1 and 4 are 1, and the values at positions 2 and 3 are 0, so the answer is 2.

For the third test case, one optimal pairing is (1, 3), (1, 5), and (5, 5). After the operations, the values at positions 1 to 5 are 0, 0, 0, 1, 0, so the answer is 1.